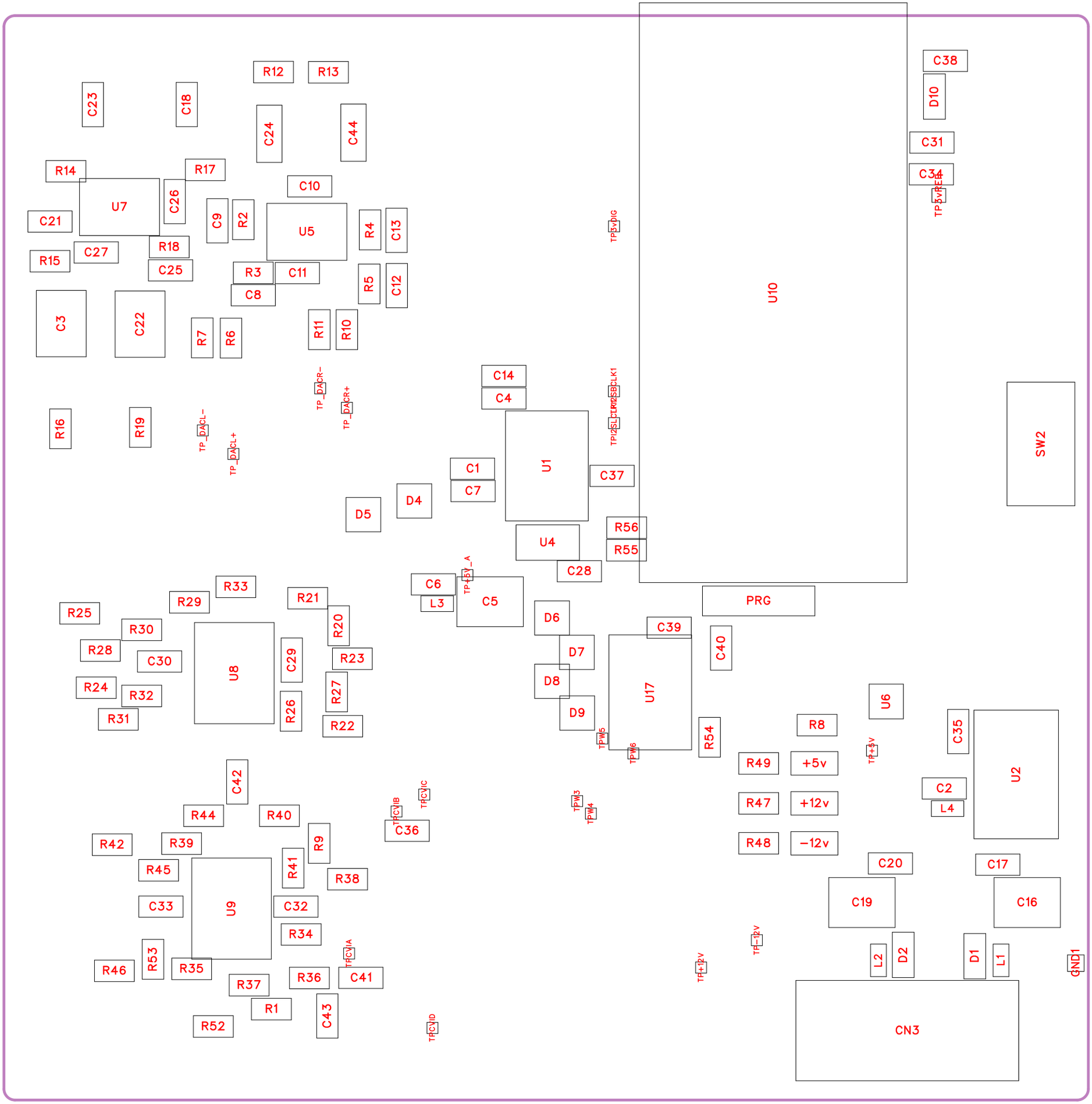
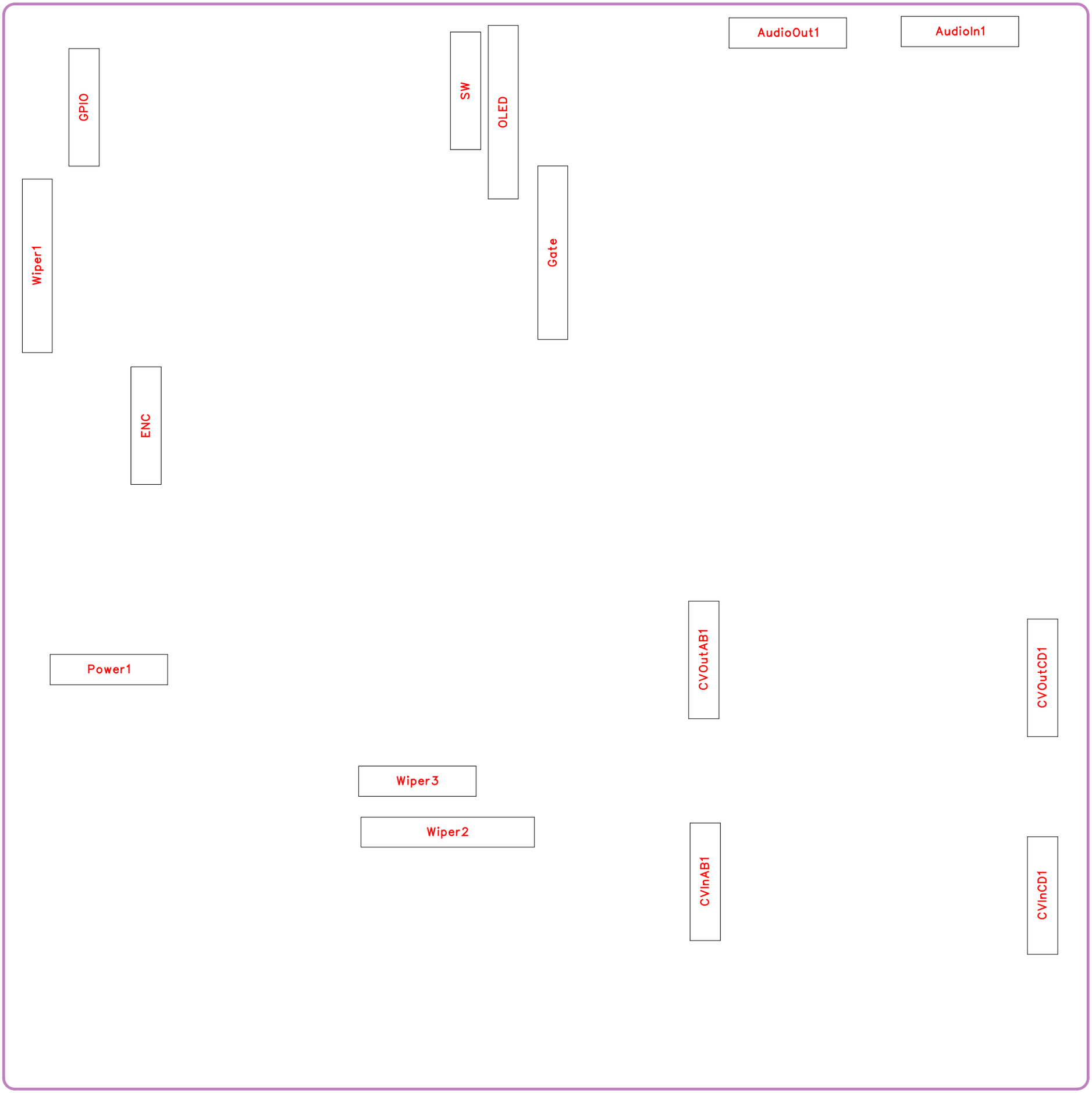
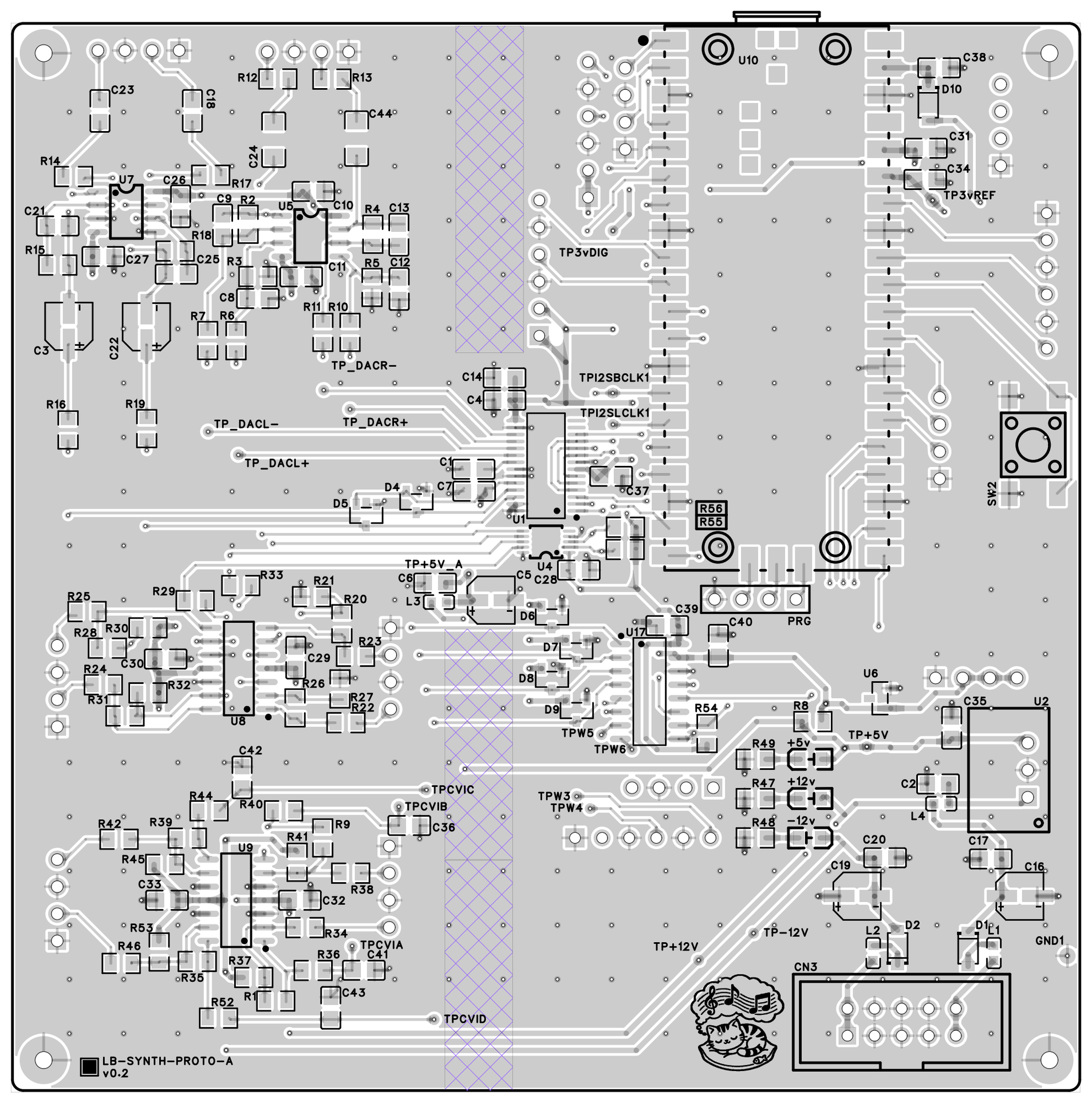

R12
R13
C23
C18
C24
C44
C38
D10
C31
C34
R14
R17
U7
C26
C10
C21
C9
R2
U5
R4
C13
R15
C27
R18
C25
R3
C11
C8
R5
C12
C3
C22
R7
R6
R11
R10
U10
C14
C4
SW2
R16
R19
TP_DACL-
TP_DACL+
TP_DACR-
TP_DACR+
C1
C7
U1
C37
D5
D4
R56
R55
U4
C28
C6
L3
C5
TP+5V_A
PRG
R33
R29
R21
R25
R30
R20
C39
C40
R28
C30
U8
C29
R23
D6
D7
D8
D9
U17
R27
R24
R32
R26
R22
R31
TPW5
TPW6
R54
U6
R8
C35
U2
R49
+5v
TP+3V
C2
L4
R47
+12v
R48
−12v
C42
R44
R40
C36
TPCVIB
TPCVIC
TPW3
TPW4
R42
R39
R9
C20
C17
R45
R41
R38
C19
C16
C33
U9
C32
R34
TPCVIA
D2
D1
L2
L1
R46
R53
R35
R36
C41
TP-12V
TP+12V
GND1
R37
R1
C43
CN3
R52
TPCVID

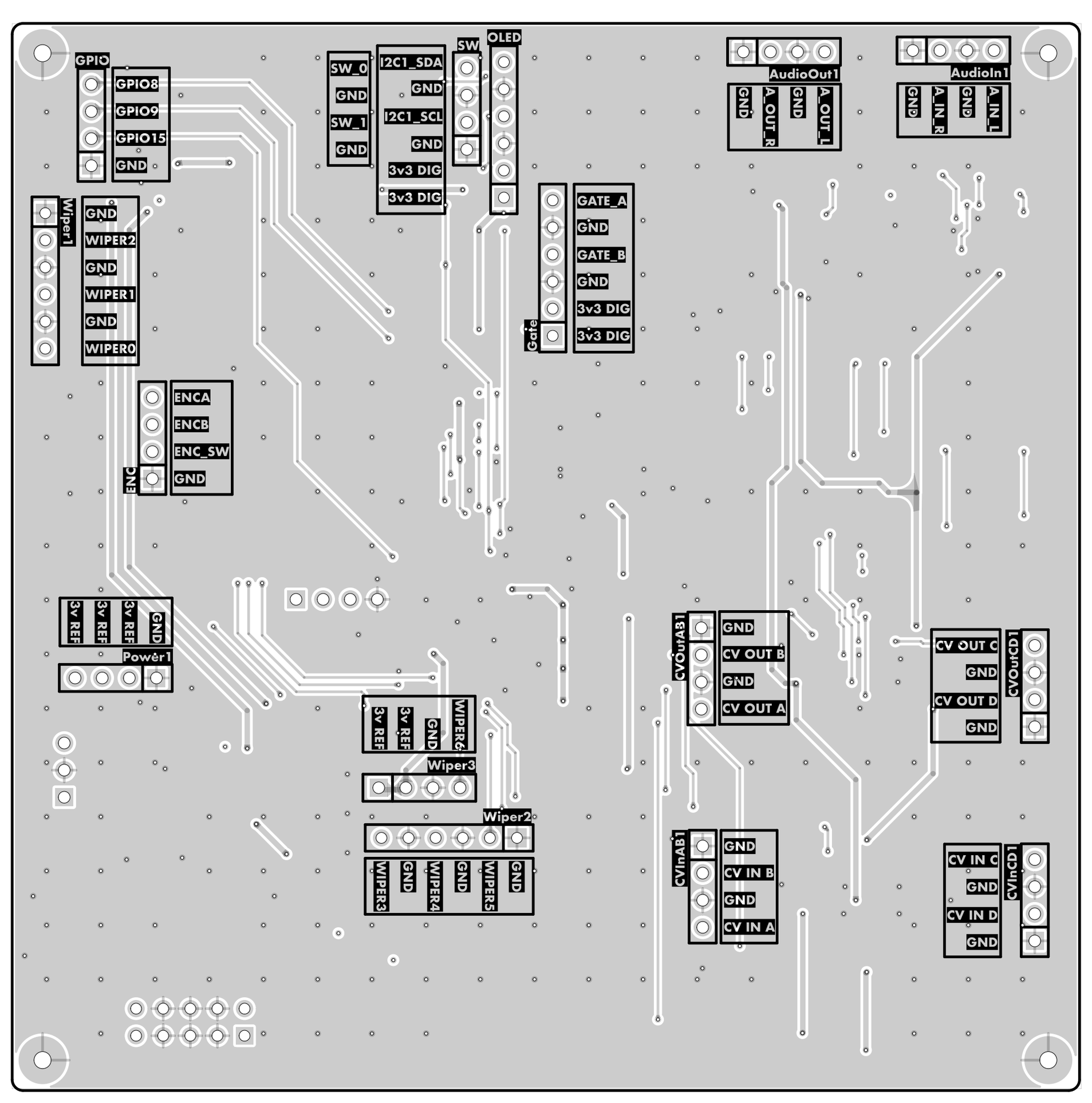

AudioOut1
AudioIn1
GPIO
SW
OLED
Wiper1
Gate
ENC
CVOutAB1
CVOutCD1
Power1
Wiper3
Wiper2
CVInAB1
CVInCD1

No.	Quantity	Comment	Designator	Footprint	Value	Manufacturer Part	Manufacturer	Supplier Part	Supplier
1	3	XL-2012SURC	+5v,+12v,-12v	LED0805-RD_RED		XL-2012SURC	XINGLIGHT(成兴光)	C965812	LCSC
2	11	2.54-1*4P母	AudioIn1,AudioOut1,CVInAB1,CVInCD1,CVOutAB1,CVOutCD1,ENC,GPIO,Power1,SW,Wiper3	HDR-TH_4P-P2.54-V-F		2.54-1*4P母	BOOMELE(博穆精密)	C2718488	LCSC
3	1	10uF	C1	C0805	10uF	CL21B106KOQNNNE	SAMSUNG(三星)	C95841	LCSC
4	1	10uF	C2	C0805	10uF	FCC0805X106K250FT	FOJAN(富捷)	C5137488	LCSC
5	5	10uF	C3,C5,C16,C19,C22	CAP-SMD_BD4.0-L4.3-W4.3-FD	10uF	RVT1V100M0405	ROQANG(容强)	C72485	LCSC
6	24	100nF	C4,C6,C7,C10,C11,C17,C20,C26,C27,C28,C29,C30,C31,C32,C33,C34,C36,C37,C38,C39,C40,C41,C42,C43	C0805	100nF	TCC0805X7R104K500DT	CCTC(三环)	C282732	LCSC
7	6	220pF	C8,C9,C12,C13,C21,C25	C0805	220pF	CC0805KRX7R9BB221	YAGEO(国巨)	C107145	LCSC
8	1	10uF	C14	C0805	10uF	CL21A106KAYNNNE	SAMSUNG(三星)	C15850	LCSC
9	2	1uF	C18,C23	C0805	1uF	CC0805KKX7R9BB105	YAGEO(国巨)	C91185	LCSC
10	2	10uF	C24,C44	C1206	10uF	CL31A106KBHNNNE	SAMSUNG(三星)	C13585	LCSC
11	1	22uF	C35	C0805	22uF	GRM21BR61E226ME44L	muRata(村田)	C86816	LCSC
12	1	IDC2.54-2*5PZ	CN3	CONN-TH_10P-P2.54_254V-10P-B		IDC2.54-2*5PZ	JXTCONN(聚兴泰)	C53207560	LCSC
13	3	1N5819HW-7-F	D1,D2,D10	SOD-123_L2.7-W1.6-LS3.7-RD		1N5819HW-7-F	DIODES(美台)	C82544	LCSC
14	6	BAT54S	D4,D5,D6,D7,D8,D9	SOT-23-3_L2.9-W1.3-P1.90-LS2.4-BR		BAT54S	R+O(宏嘉诚)	C7420333	LCSC
15	4	2.54-1*6P母	Gate,OLED,Wiper1,Wiper2	HDR-TH_6P-P2.54-V-F		2.54-1*6P母	BOOMELE(博穆精密)	C40877	LCSC
16	21	Test-Point	GND1,TP3vDIG,TP3vREF,TP+5V,TP+5V_A,TP+12V,TP-12V,TPCVIA,TPCVIB,TPCVIC,TPCVID,TPI2SBCLK1,TPI2SLCLK1,TPW3,TPW4,TPW5,TPW6,TP_DACL+,TP_DACL-,TP_DACR+,TP_DACR-	Test-Point-0.5mm					
17	4	BLM18KG601SN1D	L1,L2,L3,L4	L0603		BLM18KG601SN1D	muRata(村田)	C85833	LCSC

18	1	2.54-1*4	PRG	HDR-TH_4P-P2.54-V-M-1		2.54-1*4	ZHOURI(洲日)	C5116483	LCSC
19	4	30kΩ	R1,R9,R35,R39	R0805	30kΩ	RC0805FR-0730KL	YAGEO(国巨)	C126352	LCSC
20	4	12kΩ	R2,R3,R4,R5	R0805	12kΩ	FRC0805F1202TS	FOJAN(富捷)	C2907225	LCSC
21	5	10kΩ	R6,R7,R10,R11,R54	R0805	10kΩ	RC0805FR-0710KL	YAGEO(国巨)	C84376	LCSC
22	1	330Ω	R8	R0805	330Ω	FRC0805F3300TS	FOJAN(富捷)	C2930157	LCSC
23	13	1kΩ	R12,R13,R16,R19,R22,R23,R24,R25,R36,R40,R44,R49,R52	R0805	1kΩ	0805W8F1001T5E	UNI-ROYAL(厚声)	C17513	LCSC
24	8	100kΩ	R14,R17,R28,R31,R34,R38,R42,R46	R0805_NEW	100kΩ	RC0805FR-07100KL	YAGEO(国巨)	C96346	LCSC
25	2	22kΩ	R15,R18	R0805	22kΩ	RT0805BRD0722KL	YAGEO(国巨)	C865312	LCSC
26	2	14.3kΩ	R20,R26	R0805	14.3kΩ	FRC0805F1432TS	FOJAN(富捷)	C2933317	LCSC
27	2	10kΩ	R21,R27	R0805	10kΩ	0805W8F1002T5E	UNI-ROYAL(厚声)	C17414	LCSC
28	2	20kΩ	R29,R33	R0805	20kΩ	RC0805FR-0720KL	YAGEO(国巨)	C114566	LCSC
29	6	240kΩ	R30,R32,R37,R41,R45,R53	R0805	240kΩ	PTFR0805B240KN9	RESI(开步睿思)	C351591	LCSC
30	2	3.3kΩ	R47,R48	R0805	3.3kΩ	RMC08053.3K5%N	Tyohm(幸亚电阻)	C325790	LCSC
31	2	4.7kΩ	R55,R56	R0805	4.7kΩ	RMC08054.7K1%N	Tyohm(幸亚电阻)	C269746	LCSC
32	1	KH-6X6X5H-STM	SW2	SW-SMD_4P-L6.0-W6.0-P4.50-LS9.0_H5.0		KH-6X6X5H-STM	kinghelm(金航标)	C2837531	LCSC
33	1	PCM3060PWR	U1	TSSOP-28_L9.7-W4.4-P0.65-LS6.4-BL		PCM3060PWR	TI(德州仪器)	C2651841	LCSC
34	1	K7805-500R3	U2	PWRM-TH_3P-L11.5-W7.6-P2.54_R-785.0-0.5		K7805-500R3	YLPTEC(易川)	C2992390	LCSC
35	1	MCP4728T-E/UN	U4	MSOP-10_L3.0-W3.0-P0.50-LS4.9-BL		MCP4728T-E/UN	MICROCHIP(美国微芯)	C478093	LCSC
36	2	TL072IDR	U5,U7	SOIC-8_L5.0-W4.0-P1.27-LS6.0-BL		TL072IDR	TI(德州仪器)	C6962	LCSC
37	1	LM4040B30FTA	U6	SOT-23-3_L2.9-W1.3-P1.90-LS2.4-BR		LM4040B30FTA	DIODES(美台)	C460604	LCSC

38	2	TL074IDR	U8,U9	SOIC-14_L8.7-W3.9-P1.27-LS6.0-BL		TL074IDR	TI(德州仪器)	C6964	LCSC
39	1	Raspberry Pi Pico 2	U10	COMM-SMD_L51.0-W21.0-P2.54_SC0916		Raspberry Pi Pico 2	Raspberry Pi(树莓派)	C41407547	LCSC
40	1	MCP3208T-CI/SL	U17	SOIC-16_L9.9-W3.9-P1.27-LS6.0-BL		MCP3208T-CI/SL	MICROCHIP(美国微芯)	C626764	LCSC


C23
C18
R12
R13
C44
C24
R14
U7
C26
R17
C9
R2
U5
C10
R4
C13
C21
R18
R15
C27
C25
R3
C11
R5
C12
C8
R11
R10
R7
R6
C3
C22
TP_DACR-
C14
C4
TPI2SBCLK1
TPI2SLCLK1
R16
R19
TP_DACL-
TP_DACR+
TP_DACL+
C1
C7
D4
D5
U1
C37
U4
TP+5V_A
C6
C5
C28
L3
D6
R33
R29
R21
R20
R25
R28
R30
C30
R24
R31
R32
C29
R23
R26
R27
R22
U8
C39
C40
U17
D7
D8
D9
TPW5
TPW6
R54
R8
U6
TP+5V
R49
+5v
R47
+12v
R48
-12v
C35
U2
C2
L4
C20
C17
C19
C16
C42
R44
R40
TPCVIC
TPCVIB
C36
R42
R39
R9
R41
U9
R45
C33
C32
R38
R53
R34
R46
TPCVIA
R36
C41
R35
R37
R1
C43
R52
TPCVID
TPW3
TPW4
TP+12V
TP-12V
L2
D2
D1
L1
GND1
CN3
U10
C38
D10
C31
C34
TP3vREF
TP3vDIG
SW2
R56
R55
PRG
LB-SYNTH-PROTO-A
v0.2


GPIO
GPIO8
GPIO9
GPIO15
GND
SW_0
GND
SW_1
GND
I2C1_SDA
GND
I2C1_SCL
GND
3v3 DIG
3v3 DIG
SW
OLED
AudioOut1
GND
A_OUT_R
GND
A_OUT_L
AudioIn1
GND
A_IN_R
GND
A_IN_L
Wiper1
GND
WIPER2
GND
WIPER1
GND
WIPER0
GATE_A
GND
GATE_B
GND
3v3 DIG
3v3 DIG
Gate
ENC
ENCA
ENCB
ENC_SW
GND
3v REF
3v REF
3v REF
GND
Power1
3v REF
3v REF
GND
WIPER6
Wiper3
Wiper2
WIPER3
GND
WIPER4
GND
WIPER5
GND
CVOutAB1
GND
CV OUT B
GND
CV OUT A
CV OUT C
GND
CV OUT D
GND
CVOutCD1
CVInAB1
GND
CV IN B
GND
CV IN A
CV IN C
GND
CV IN D
GND
CVInCD1

